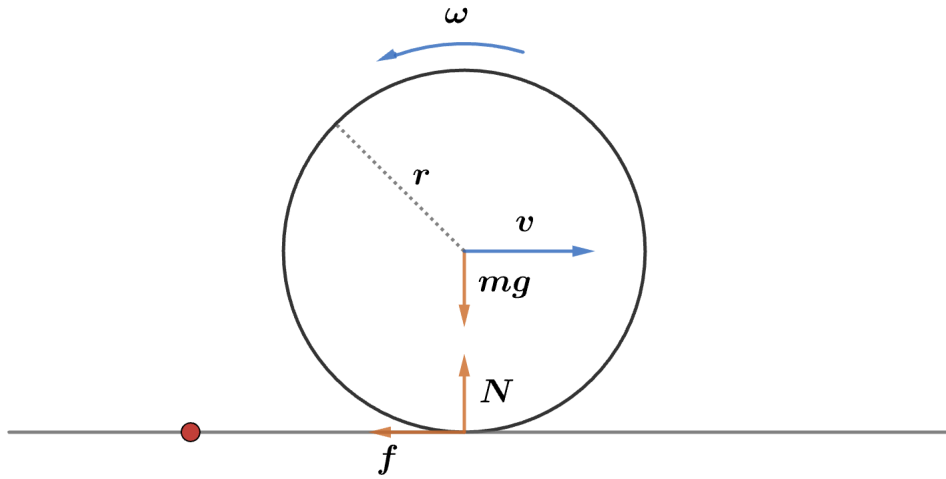


2016 F=ma Exam: Problem 17

Kevin S. Huang



Choosing a point on the ground as the pivot point, note that there is no net torque on the ping pong ball. The line of the friction force goes through the pivot point and the normal force cancels out gravity. Thus, we have conservation of angular momentum. If the ball stops in the end then $L_f = 0$. The initial angular momentum is

$$L_i = mvr - I\omega$$

where $I = \frac{2}{3}mr^2$ is the moment of inertia of a spherical shell. Since $L_i = L_f$,

$$mvr = I\omega = \frac{2}{3}mr^2\omega$$

$$v = \frac{2}{3}r\omega$$

so the answer is A.